

Video anomaly detection system using deep convolutional and recurrent models

Maryam Qasim^{a,*}, Elena Verdu^b

^a Department of Law, Shaheed Benazir Bhutto University, Sheringal, Dir, KPK, Pakistan

^b Universidad Internacional de La Rioja, Logroño, Spain

ARTICLE INFO

Keywords:

Anomaly detection

Video surveillance

Deep learning

CNN

Simple recurrent unit (SRU)

ResNet

UCF-Crime

ABSTRACT

Automatic identification of anomalies in video surveillance is an interesting research field. Even though interactive multimedia anomaly detection algorithms have been developed, it is still hard for video surveillance to find unusual things like illegal activities and crimes. In this study, a deep convolutional neural network (CNN) and a simple recurrent unit (SRU) are used to build an automated system that can find anomalies in videos. The ResNet architecture takes high-level feature representations from the video frames that come in, while the SRU collects temporal features. The SRU has expressive recurrence and allows for highly parallelized implementation, which makes the video anomaly detection system more accurate. In the study, three models to detect anomalies are suggested as ResNet18 + SRU, ResNet34 + SRU, and ResNet50 + SRU, respectively. The suggested models are examined using the UCF-Crime dataset. This study made a clear distinction between normal and unusual actions, showing that CNN + SRU were able to put each unusual action in the right category. Using the UCF-Crime dataset, ResNet18 + SRU achieved 88.92% accuracy, ResNet34 + SRU achieved 89.34% accuracy, and ResNet50 + SRU achieved 91.24% accuracy. Furthermore, the proposed models demonstrated significantly higher performance accuracy and outscored the comparable deep learning models.

1. Introduction

Both the business world and academia find video surveillance to be an important and exciting research topic [1,2]. Due to the growing safety concerns [3] in many areas, including train stations, airports, military sites, and retail centers, video surveillance is receiving a lot of interest in the research community. Since it's hard to make detection techniques without knowing about the target activity and the scene, it's important to find out if suspicious things are happening there [3]. In order to locate and identify anomalies in a complex environment, video surveillance is mostly focused on crowded situations. An anomaly is an unusual event or an irregular occurrence. The effort of modeling and processing the anomalous scene outcomes might be difficult and might even seem impossible [4,5]. The identification of anomalies in video surveillance is a key activity that may be challenging since anomalies are defined as behaviors that deviate from the established patterns. An anomalous activity may be a typical activity in certain settings, such as a shooting at a gun club. In shooting clubs, shooting is normal despite the fact that it is often an abnormal behavior. As an alternative, certain behaviors may not necessarily be abnormal but still be anomalous under

certain settings [6,7].

Killing, looting, molestation, and intense assaults are only a few examples of the abnormal behaviors that are often noted. Killing is the intentional taking of another person's life. Looting is the act of taking someone else's possessions by employing severe physical force and violence. Sexual exploitation of individuals (men, women, and children) against their will is molestation. This unlawful action is horrible and has serious repercussions. Intensive assaults are unlawful conflicts between people that are carried out to get something or cause damage to others [8]. A system that grabs notice is anomaly detection and prevention utilizing deep learning. To protect public safety, several departments of law enforcement throughout the world are experimenting with deep learning (DL) systems. A law enforcement department may learn from the predicted anomaly operations and large volume of data processing that reveal anomalous patterns. Anomaly activities can go unnoticed because of external influences coming from all sectors of society. Because of this, an intelligent security system may identify anomalous activity on its own and aid in excluding manipulative activity by alerting law enforcement agencies and avoiding people. A case study of Natal, Brazil and San Francisco, USA, where anomalous activity has been

* Corresponding author.

E-mail address: maryam@sbbu.edu.pk (M. Qasim).

<https://doi.org/10.1016/j.rineng.2023.101026>

Received 21 October 2022; Received in revised form 15 February 2023; Accepted 26 February 2023

Available online 23 March 2023

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prevalent and is being watched by sophisticated video surveillance systems [9].

In order to extract crucial information from recorded videos, computer vision automates the identification of anomalous behaviors by alerting a system that enforces the law when anomalous activities are observed [10,11]. A pattern of anomalous behavior may be discernible by carefully examining various anomalous behaviors while entering or leaving public spaces [12]. In a select instance, historical patterns may identify malicious people in the captured videos [13,14]. When abnormalities are noticed, the feedback can automate the processes to extract data from recorded videos using DL models [15]. From the standpoint of DL, the classification of anomalous activities is divided into supervised, unsupervised, and semi-supervised learning models. One deep learning approach trains the model on both typical and atypical behaviors [16]. On the other hand, in a multi-model learning context, deep learning models are trained using both normal and anomalous activity [17]. Several research studies used supervised deep learning to find anomalous behaviors in videos [18–25]. For anomaly detection, a variety of DL models are used, including Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long-Short Term Memory (LSTM), Gated Recurrent Units (GRUs), Simple Recurrent Unit (SRUs), and Generative Adversarial Networks (GANs) [26–30].

This paper suggests leveraging a combined deep learning model to identify anomalous behavior from a multiple-learning approach. Numerous anomalous behaviors in the real world are classified as anomalies. However, the emphasis of this work is on anomalous behaviors identified in the UCF-Crime dataset [31], which includes abnormal and violent behavior caught on camera in a variety of public locations. The ResNet-18/34/50 has been used as a CNN model in the proposed system for anomaly detection to extract high-level data from video frames. To identify temporal relationships in the video dataset, the SRU, RNN model is supplied with spatial features extracted from the CNNs and presents an expressive recurrence and enables highly parallelized implementation. The proposed system generates output that indicates if the input video includes anomalous or normal behavior. This proposed model can increase the precision of abnormal activity detection and lessen the restrictions of human-monitored video surveillance systems. The CNN + SRU also significantly speed up the response time. A compact neural network for anomaly detection is provided by using a SRU to learn the temporal features in video frames. The number of network parameters is drastically decreased by a convolutional layer as compared to the fully connected layer in the standard RNNs (LSTM/GRU). Various kinds of anomalous occurrences from the UCF-Crime dataset are used to assess the proposed anomaly detection system.

The remainder of this study is structured as follows: Section 2 provides a brief description of the related studies on anomaly detection modeling using DL in the literature. Section 3 presents the details of the ResNet + SRU anomaly detection system, and Section 4 reports the experiments that were conducted to compare the proposed system with competing systems, and analyzes the results. Lastly, Section 5 presents the conclusion of this study.

2. Related literature study

An anomaly detection system's purpose is to predict and prevent anomalous (criminal) actions. Although traditional, non-deep learning techniques are helpful, yet they act on their own. As a result, a machine capable of integrating the key components of traditional methodologies would be tremendously beneficial. Using the Waikato Environment for Knowledge Analysis (WEKA) platform [29], investigated violent crime trends between the community's dataset and true criminal statistics data. Three models are used which are linear regression, additive regression, and decision stump. The linear regression on randomly selected samples in testing was able to manage unpredictability, demonstrating improved detection across models and demonstrating the

effectiveness of deep learning in recognizing violent trends and criminal tendencies.

A study by Ref. [32] examined the anomaly detection in urban areas where anomalies have been combined into a grid size 200×250 m and observed with hindsight. A model of ensemble logistic regression and neural networks is proposed to detect anomalies. The results indicate that fortnight predictions improve extraordinarily as compare to the monthly predictions. Anomaly activities are identified and observed in Ref. [33] utilizing anomaly data for the last 15 years from the Vancouver city. The KNN and decision tree detected anomaly activities. The samples sets of 560,000 anomaly dataset were examined and anomaly activities were predicted with accuracy of 39%–44%. Another study [34] predicted anomaly statistics in Philadelphia to determine anomaly trends. ML algorithms (KNN, logistic regression, Ordinal regression, and decision trees) were used to train for getting anomaly predictions with large datasets. The models were able to determine the trend of anomaly activity with 69% accuracy. Models using data science were applied to detect the anomalies in the Chicago criminal dataset. Various ML models (SVM, logistic regression, decision trees, KNN, random forest, and Bayesian models) were tested and nominated as precise models for training. The best accuracy (78.7%) was achieved with the KNN. This study suggested that law enforcement agencies use deep learning models to predict, find, and solve unusual activities more quickly. This would help reduce crime in the public.

A GUI-based deep learning model for predicting the anomalies is proposed in Ref. [35]. The results of supervised models are compared to predicted anomalies. A feature-level data fusion-based DNN with environmental context information is presented to identify anomalies with high accuracy utilizing cross-model data from multiple domains [36]. An online crime statistics database was used for the model training including SVM, regression, and kernel density estimation). The SVM and KDE obtained 67.01% and 66.33% accuracy, respectively, whereas the proposed model obtained an accuracy of 84.25%. A study by Ref. [5] used previous crime locations to predict anomalies likely to happen in old locations. To examine and understand the data, the Levenberg-Marquardt and scaled techniques were applied to evaluate the data where the scaled technique outperformed. The ANOVA verified that with scaled technique 78% crime rates were reduced showing an accuracy of 78%.

The study in Ref. [37] offered a methodology for predicting anomalies by studying a dataset of previously committed anomalies and their trends. KNN, decision tree with adaptive boosting, and random forest were used to improve prediction accuracy. The statistics are classified as uncommon or frequent. Over a period of 12 years, the deep learning architecture was trained on criminal activity in San Francisco, California where 99.2% accuracy was attained by using oversampling and under-sampling with random forest. Other studies have achieved state-of-the-art results for crime detection in various studies [38–46]. Various CNN types have been formulated, such as the AlexNet, ResNets, VGG, Inceptions, and their variants. Many studies combined these CNNs with a softmax layer [38] and morphological analysis [39] to detect anomalies. Besides CNN [40,41], offered autoencoders. A Bayesian nonparametric [42] is proposed to detect abnormal events in videos. Since surveillance camera feeds are sequential data, the LSTMs have gained attraction for anomaly detection. In an unsupervised learning fashion, encoder-decoder LSTM [43] is proposed. Spatiotemporal networks (STNs) are gaining popularity to learn spatial and temporal features [44] where RNNs and CNNs jointly extract spatiotemporal features for anomaly detection. Another model where a convolutional layer filters the output of CNNs before feeding an LSTM is called the ConvLSTM [45]. An alternative to the fully connected layer in LSTM, a convolutional layer intensely reduces the number of parameters. The GRU further reduce the parameters if replaced with LSTM in ConvLSTM. There is a 25% reduction in the parameters. There are very limited studies implemented by ConvGRU to detect anomalies in video streams. Table 1 summarizes the previous work with achieved accuracy.

Table 1

Summary of Previous work on Crime detection.

S. No.	Reference	Neural Model	Achieved Accuracy
1	[36]	Decision Tree	59.15%
2	[37]	KNN	87.03%
3	[38]	Naive Bayes	87.00%
4	[39]	ARIMA	86.00%
5	[40]	Regression Model	72.00%
6	[41]	SVM	84.30%
7	[42]	Random Forrest	97.00%
8	[43]	GoogleNet	87.16%
9	[44]	GBM	88.00%
10	[46]	BO-PSM	88.00%

3. Proposed video anomaly detection system

Fig. 1 shows the diagram of the proposed video anomaly detection system in videos. To extract features from DNNs, residual networks (ResNets) are efficient neural models [46]. In order to extract spatial information from the input video frames, ResNet must first be deployed. The next step is to extract the temporal dependencies in video frames using SRU as an RNN model. As inputs for ResNet, the videos are split up into sequences of m frames, and the SRU is then fed the outputs of the ResNet. To find the anomaly in video frames, the spatiotemporal attributes are processed through maxpooling and fully-connected layers. The ResNet-18, ResNet-34, and ResNet-50 have been used as CNN models in the proposed system for anomaly detection to extract high-level data from the video frames.

3.1. Video pre-processing

The proposed CNN + SRU system for anomaly detection in videos is shown in Fig. 1, where input video is first preprocessed and segmented into fixed frames with a frame rate of 30 frames per second. As a result, there are 1800 total frames in a 60-s video. After selecting the frames, the difference between each frame and its neighboring frame is computed in order to take all the input frames' spatial motions into account. We chose three categories from the UCF-Crime dataset. Each video's specific period of anomalous activity was divided into two halves and labeled as an anomaly; the remaining video was given the label normal. The videos are then separated into segments of equal duration. Thus, from m frames, k frames are chosen. Only anomalous activities are thus highlighted. Additionally, the regular activities are chosen from the identical videos that include the abnormal activities. Other than actions, every setting is the same as it was in the UCF-Crime dataset. Such configurations improve the system's ability to identify anomalous activity. Full training videos incur significant computational expense. As a result, we have taken into consideration a training framework over a predetermined number of frames, namely, a targeted bag that includes the key information required to interpret the actions in

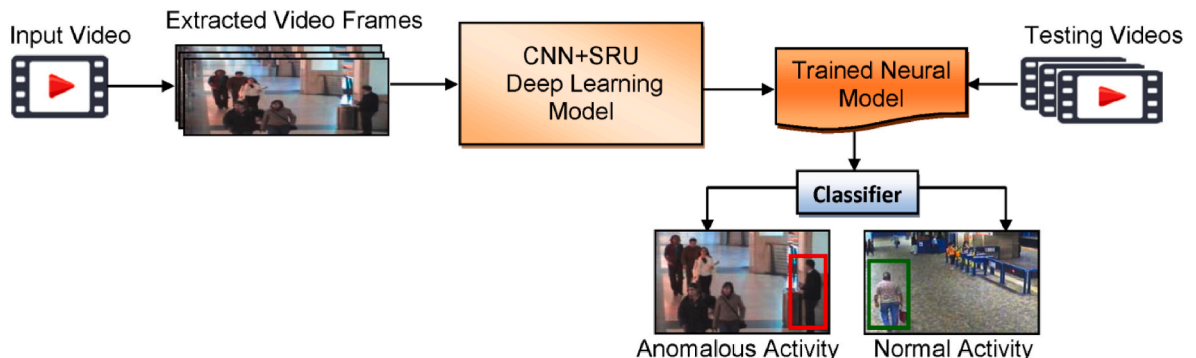
the videos, followed by block formulation and selection. A targeted bag is a group of frames made up of actions in a full-length recorded video. The extraction of this targeted bag is shown in Fig. 2, where just a small proportion of the complete recorded video is identified as suspicious or criminal actions. With Z-frames from W-frames, they are thus seen as a targeted bag. By using this method on every video feed in the database, the amount of training data was greatly cut down by getting rid of duplicate information.

3.2. ResNet CNN models

On several common datasets, including ImageNet, ResNet has shown outstanding performance. There are several variations of ResNets, including ResNet-18, ResNet-34, ResNet-50, ResNet-101, and ResNet-152. However, CNN + SRU have been implemented on ResNet-18, ResNet-34, and ResNet-50 because of its superior design and performance for our tasks. The proposed approach employs Transfer Learning [47,48] to prevent problems with labeling anomalies, and CNN + SRU is thus pre-trained using the ImageNet dataset, which consists of 1000 sets of images. The model parameters are loaded and updated by running ResNet on ImageNet, making it ready to run on the desired datasets. The ResNet can process $(240 \times 240 \times 3)$ dimension data due to the input frame size of (240×240) . A 4-D tensor $(n \times 1 \times 1 \times 2048)$ output is created from the Deep Residual Features (DRF) while passing through the convolutional and pooling layers, which is then reshaped before being fed to the SRUs. Fig. 3 shows the ResNet-18, ResNet-34, and ResNet-50 structures, and Table 2 contains the architecture of three CNN models. The outputs of ResNet are transformed into $(n \times 3 \times 3 \times 128)$ and given to the SRU layers. A fully connected dense layer is not used since ResNet does not use it for categorization. Due to improved parallelization and gradient propagation, the parts work together to create a basic yet powerful design that is also simple to scale.

3.3. SRU layer

Due to the inherent difficulties of parallelizing their state calculations, recurrent neural networks (RNNs) scale poorly. The Simple Recurrent Unit (SRU) [49] in this study is a lightweight recurrent unit that establishes a balance between model capacity and scalability. For the purpose of facilitating the training of deep models, SRU is built to provide expressive recurrence, allow highly parallelized implementation, and include rigorous initialization. We have applied to the task of anomaly detection in video frames to show the efficiency of SRU when combined with CNNs. On the classification datasets, we assume that SRU outperforms LSTM and convolutional models and is faster than optimized LSTM. Incorporating SRU into the design also allows us to translate data more efficiently than the Transformer model. This section introduces and describes the SRU design. The following calculations are involved in a single layer of SRU:

**Fig. 1.** Proposed anomaly detection framework.

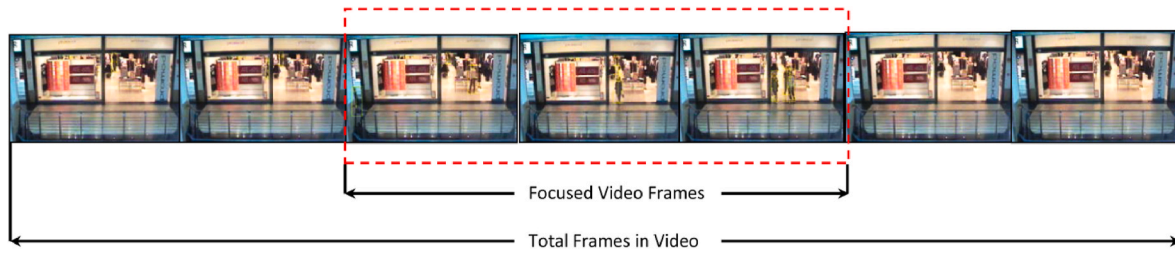


Fig. 2. Focused Bag frames extraction.

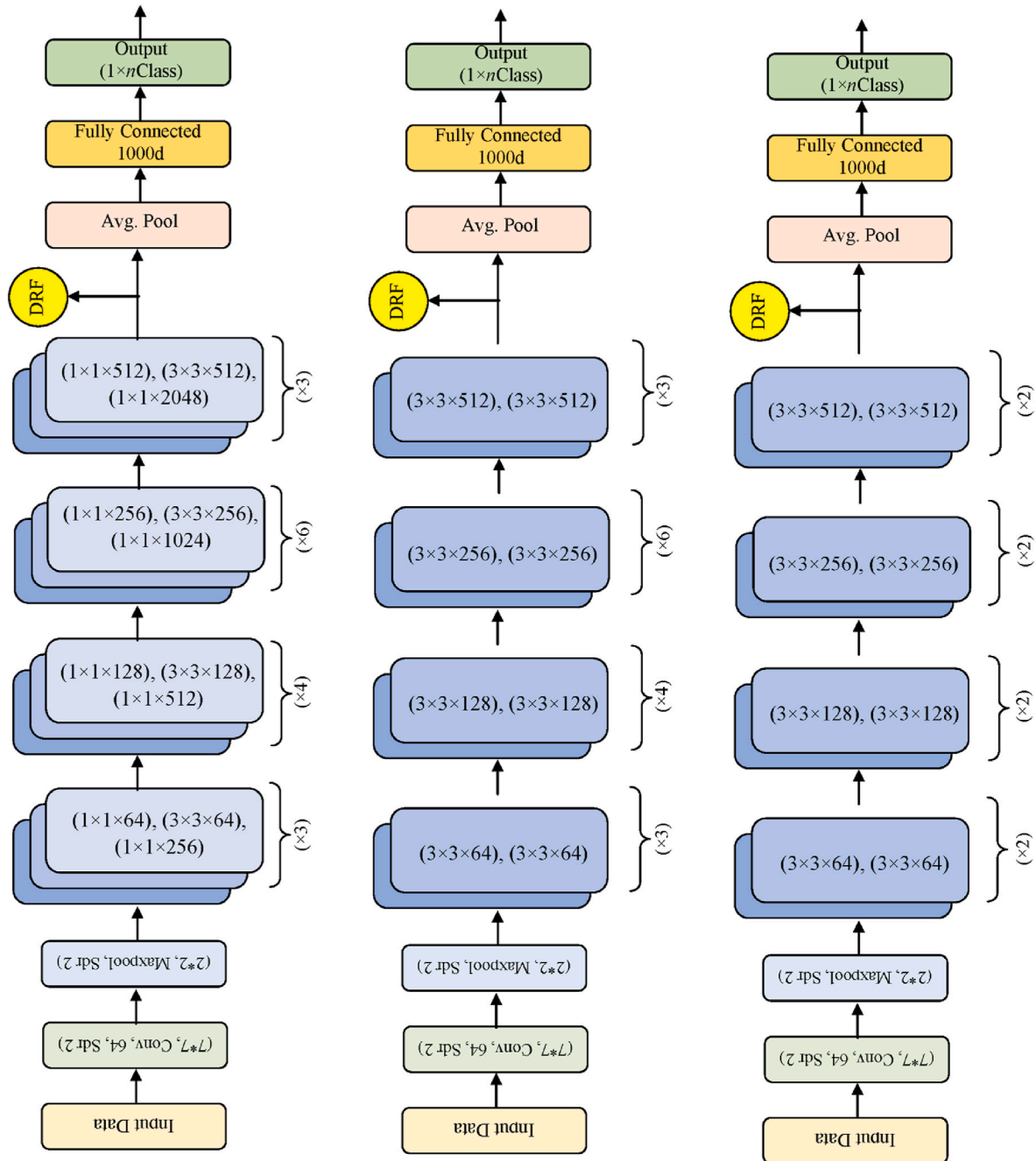


Fig. 3. ResNet architecture: ResNet-50 (left), ResNet-34 (middle), ResNet-18 (right).

$$f_t = \sigma(\mathbf{W}_f x_t) + V_f \odot C_{t-1} + b_f$$

(1)

$$r_t = \sigma(\mathbf{W}_r x_t) + V_r \odot C_{t-1} + b_r$$

(2)

Table-2
ResNet-50/34/18 architectures.

Layer Name	Output Size	50-Layers Model	34-Layers Model	18-Layers Model
Conv1	(112 × 112)	(7 × 7), 64, Stride 2 (3 × 3), Maxpool, Stride 2	(7 × 7), 64, Stride 2 (3 × 3), Maxpool, Stride 2	(7 × 7), 64, Stride 2 (3 × 3), Maxpool, Stride 2
Conv2	(56 × 56)	[(1 × 1, 64), (3 × 3, 64), (1 × 1, 256)] × 3	[(3 × 3, 64), (3 × 3, 64)] × 3	[(3 × 3, 64), (3 × 3, 64)] × 2
Conv3	(28 × 28)	[(1 × 1, 128), (3 × 3, 128), (1 × 1, 512)] × 4	[(3 × 3, 128), (3 × 3, 128)] × 4	[(3 × 3, 128), (3 × 3, 128)] × 2
Conv4	(14 × 14)	[(1 × 1, 256), (3 × 3, 256), (1 × 1, 1024)] × 6	[(3 × 3, 256), (3 × 3, 256)] × 6	[(3 × 3, 256), (3 × 3, 256)] × 2
Conv5	(7 × 7)	[(1 × 1, 512), (3 × 3, 512), (1 × 1, 2048)] × 3	[(3 × 3, 512), (3 × 3, 512)] × 3	[(3 × 3, 512), (3 × 3, 512)] × 2
	(1 × 1)	Average Pool, 1000d Fully Conn., Softmax		
FLOPS		3.8 × 10 ⁹	3.6 × 10 ⁹	1.8 × 10 ⁹

$$c_t = f_t \odot C_{t-1} + (1 - f_t) \odot Wx_r \quad (3)$$

$$h_t = r_t \odot C_t + (1 - r_t) \odot x_t \quad (4)$$

where W , W_f and W_r are parameter matrices, whereas V_f , V_r , bf , and b_r are parameters to be learnt during training. The whole network design could be separated into two sections: a highway network (Eqs. (3) and (4)) and a light recurrence (Eqs. (1) and (2)). The light recurrence section computes the sequence of states c_t while reading the input vectors x_t one at a time in order to collect sequential data. Gradient-based deep network training is made easier by the highway network component. The input x_t and the state c_t generated by the light recurrence (Eq. (4)) are adaptively combined using the reset gate r_t (Eq. (3)), where $(1 - r_t) \odot x_t$ is a skip connection that enables the gradient to immediately propagate to the preceding layer. It has been demonstrated that these connections maximize scalability. The SRU internal structure is depicted in Fig. 4.

4. Experiments

4.1. Dataset and evaluation measures

In this paper, the proposed model is implemented on the UCF-Crime dataset [31] which includes abnormal, illegal, and violent behavior recorded by surveillance video cameras located in public places such as stores and streets. The UCF-Crime dataset is prepared from everyday

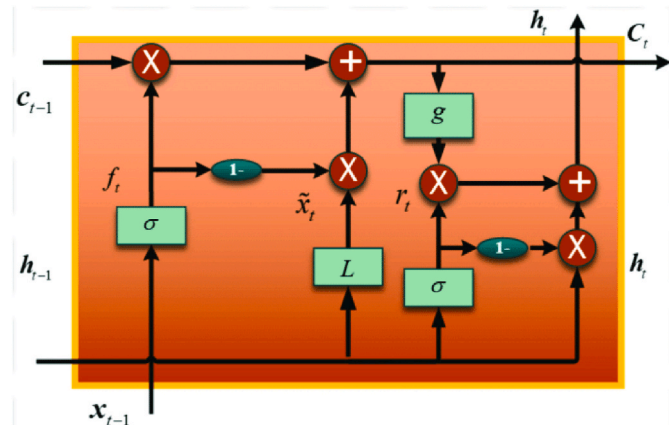


Fig. 4. Internal structure of SRU.

actual events, which is the key reason for selecting this dataset. Many studies have relied on handicraft datasets or specific datasets with similar backgrounds and environments (for example, fighting and movie datasets), which are unrelated to our daily lives. The UCF-Crime dataset includes lengthy surveillance video camera feeds covering 13 different classes of anomaly events such as abuse, arrest, arson, assault, road accidents, burglaries, explosions, fighting, robbery, shooting, stealing, shoplifting, and vandalism in addition to the normal events class. For comparison with other related studies, the training and testing data is arranged as 75% and 25% in experiments. The example samples of the UCF-Crime dataset are given in Fig. 5. Two variants of the UCF-Crime dataset, including Ucfcrimes and Binary, are used in the experiments, where Ucfcrimes contains 14 classes whereas Binary has 13 abnormal events. The quantity of videos for each class from the UCF-Crime and Binary datasets is given in Table 3. The model is evaluated for accuracy, precision, F1, and AUC, respectively.

4.2. Model setting and selection

In experiments, the proposed model is applied by using ResNet-18, ResNet-34, and ResNet-50, which are available in the Keras library. To tune the model, several hyperparameters are used to attain the best performance. Table 4 shows the results of experiments with different types of weight initialization and optimizers. As a result, to initialize the model weights, glorot-uniform (Xavier) is utilized, whereas to optimize the model, the RMSprop optimizer is imposed. The learning rate and the number of epochs are fixed at 0.0001 and 100, respectively. However, an early stop is applied when the loss converges. The video sequence length is fixed at 20 frames. Since then, a focused bag has been used for video frames. Table 5 provides a comparison of total video frames and frames in the focus bag. In experiments, different kinds of evaluation are used. During the first step, CNN-SRU is tested with several CNN models such as InceptionV3, VGG19, ResNet-18, ResNet-34, and ResNet-50 available in the Keras library. Table 6 shows a comparison in terms of accuracy. Keras and Tensor flow platforms are used for simulations.

5. Results and discussions

First, the proposed ConvGRU-CNN model is examined by measuring its accuracy (Acc), precision (Prc), and F1-scores. Table 7 provides the Acc, Prc, and F1 scores for ResNet50 + SRU, ResNet34 + SRU, and ResNet18 + SRU, respectively. It is clear from Table 7 that the proposed CNN-SRU models achieved significant measuring scores for 14 categories of the UCF-Crime dataset. The measuring results are averaged over 14 anomaly activities. Significant average accuracies (88.89%, 89.94%, and 91.44%) are achieved with the proposed ResNet18 + SRU, ResNet34 + SRU, and ResNet50 + SRU models for crime detection, with the UCF-Crime and binary datasets. A higher accuracy is obtained with ResNet50 + SRU, which is 1.5% and 2.55% higher than ResNet34 + SRU and ResNet18 + SRU. In addition, better precision and F1 are achieved with a large number of the anomaly categories. The proposed models attained encouraging average precision, F1-scores, and AUC (ResNet18 + SRU: 89.69%, 88.89%, and 89.08%), (ResNet34 + SRU: 90.46%, 90.0%, and 90.09%), and (ResNet50 + SRU: 91.44%, 91.71%, and 91.64%).

5.1. Comparison with other models

In experiments, the proposed CNN + SRU models are compared with other CNN models by measuring the accuracy (Acc) and area under the curve (AUC). Limited literature on anomaly detection by using the UCF-Crime dataset is available. Table 8 provides the AUC scores for the binary classification on the UCF-Crime dataset for the proposed model and other neural models for anomaly detection. The related models include support vector machine (SVM) [45], MIL [31], 3D-CNN [12], TSN [44], AutoEncd [41], SCL [50], CNN-RNN [51], and UGD-KM [52]. The

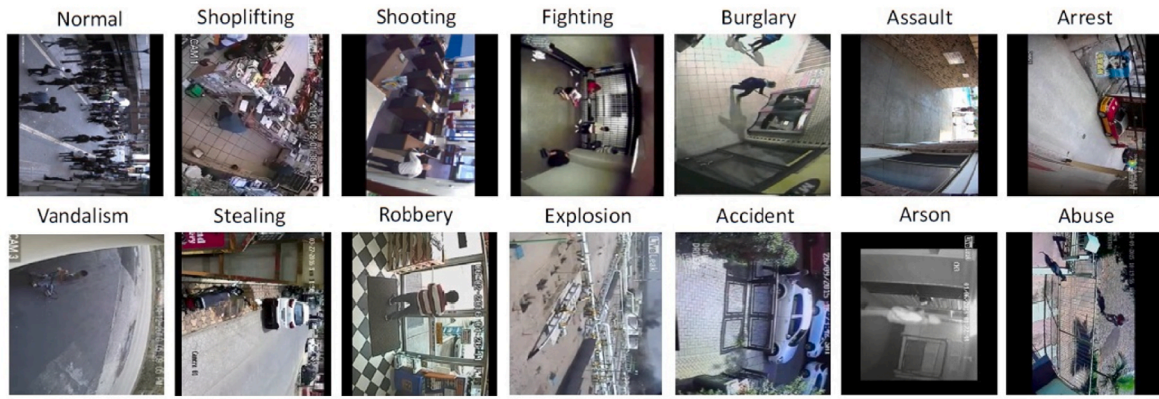


Fig. 5. The example samples of the UCF-Crime dataset.

Table 3

Number of videos for UCF-Crimes and Binary datasets.

Anomalies	Videos (Ucfcimes)	Videos (Binary)	Anomalies	Videos (Ucfcimes)	Videos (Binary)
Abuse	50	50	Road Accident	50	150
Arrest	50	50	Robbery	50	150
Arson	50	50	Shooting	50	50
Assault	50	50	Shoplifting	50	50
Burglary	50	100	Stealing	50	100
Explosion	50	50	Vandalism	50	50
Fighting	50	50	Normal	50	950
Total	350	400	Total	350	1500

Table 4

Hyperparameters settings.

Hyper Parameters	Tuning	Accuracy (%)
Weights Initialization	Glorot-Uniform (Xavier)	91.11%
Weights Initialization	Random-Uniform	89.23%
Weights Initialization	He-Uniform	89.07%
Optimizer	Adam	91.43%
Optimizer	RMSprop	90.30%

Table 5

Video frames analysis (in example anomalies).

Anomaly	Total Video Frames	Focused Video Frames
Assault	130,395	55,023
Fighting	269,255	132,517
Shooting	157,735	55,427
Vandalism	157,511	82,163
Total	714,896	156,610

Table 6

CNN Models comparison.

Hyper Parameters	Weights Initialization	Accuracy (%)
ResNet18 + SRU	Glorot-Uniform (Xavier)	88.23%
ResNet34 + SRU		89.57%
ResNet50 + SRU		91.12%
InceptionV3+DRU	RMSprop	86.33%
VGG19 + SRU		88.87%

categories for all the normal events mentioned above are considered the anomaly categories, whereas data with no abnormal events is considered normal. The testing classifier indicates the probabilities of correctly classified anomaly events. Table 8 shows that the proposed CNN + SRU

Table 7

Model Evaluation in terms of Acc, Prc, and F1 scores.

Models	Database	Accuracy	Precision	F1	AUC
ResNet18 + SRU	Ucfcimes	88.65%	89.51%	88.66%	88.71%
	Binary	89.12%	89.87%	89.11%	89.44%
	Average	88.89%	89.69%	88.89%	89.08%
ResNet34 + SRU	Ucfcimes	89.84%	90.35%	89.75%	89.87%
	Binary	90.03%	90.56%	90.24%	90.31%
	Average	89.94%	90.46%	90.00%	90.09%
ResNet50 + SRU	Ucfcimes	91.25%	91.54%	91.94%	91.74%
	Binary	91.63%	91.88%	91.33%	91.53%
	Average	91.44%	91.71%	91.64%	91.64%

Table 8

Model comparison in terms of AUC scores.

Reference	Models	AUC (%)
[45]	SVM	50.10%
[39]	AutoEncd	50.60%
[31]	MIL (Loss with No Constraints)	74.44%
[31]	MIL (Loss with Constraints)	75.41%
[42]	TSN	78.08%
[29]	3D-CNN	81.01%
[47]	CNN-RNN	81.77%
[48]	UGD-KM	64.30%
ResNet18 + SRU (Proposed)	CNN + SRU	89.08%
ResNet18 + SRU (Proposed)	CNN + SRU	90.09%
ResNet18 + SRU (Proposed)	CNN + SRU	91.64%

models outperformed the related benchmark models in anomaly detection. For example, the AUC score improved from 50% with SVM to 89.08% with ResNet18 + SRU and achieved a 38.98% AUC gain. Similarly, AUC with AutoEncd is improved from 50.60% to 90.09% with ResNet34 + SRU achieving a large performance gain of 39.49%. Similarly, the AUC improved from 75.41% with MIL to 91.64% with ResNet50 + SRU. On average, the highest AUC is achieved by the ResNet50 + SRU model. Fig. 6 shows a performance improvement over competing models. In comparison to 3D-CNN, the proposed CNN + SRU improved the AUC from 81.05% to 91.64%, whereas with MIL, the UGD-KM improved by 27.34%.

5.2. Ablation study

To observe the efficiency of CNN + SRU for crime detection, the CNN + LSTM and CNN + GRU were implemented for crime detection and the accuracy as well as the AUC of three models was compared as an ablation study. This set of experiments was performed on the same experimental settings as done for CNN + SRU. Table 9 shows the results of the ablation study. The results indicate that CNN + LSTM

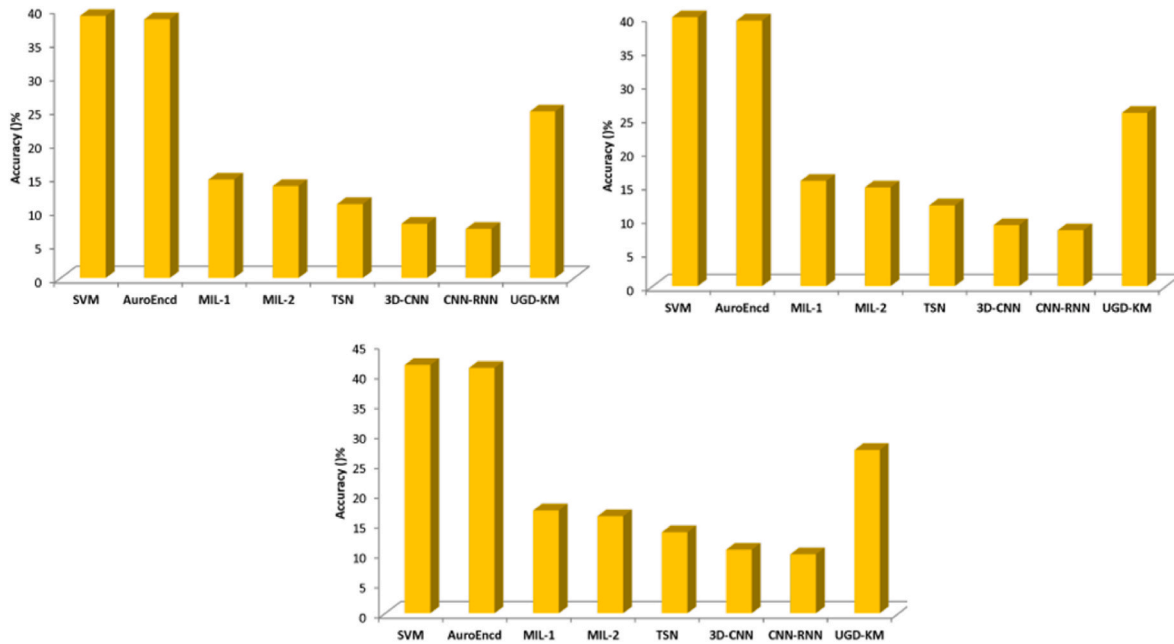


Fig. 6. Improvements over other models, ResNet-18 (Upper Left), ResNet-34 (Upper Right), ResNet-50 (Bottom).

Table 9

Ablation in terms of Accuracy, AUC and Computational cost.

Database	Accuracy	AUC	Computational Cost
CNN + LSTM	88.65%	88.66%	25% and 11.2% less computational cost with CNN + SRU as compared to CNN + LSTM and CNN + GRU
CNN + GRU	89.12%	89.11%	
CNN + SRU	90.19%	90.20%	

underperforms in terms of accuracy, AUC, and computational cost, respectively. The proposed CNN + SRU method obtained better results with less computational cost.

6. Laws preventing the anomaly/criminal activities in Pakistan

Anomaly actions, including violent events and malicious prosecution, have grown in various south Asian countries such as Pakistan and throughout the world when inflation has escalated. It is critical to develop new, effective methods of prevention. Traditional methods of detecting anomalous patterns are still in use, although technology has advanced. As a result, it is advised that Artificial Intelligence (AI) be used to deter criminal incidents from occurring. The governments and other relevant institutions are in charge of preventing and decreasing crime. As a consequence, one of the key answers is contemporary technology. To address this problem, legislative authorities adopted the Prevention of Electronic and Other Crimes Act, which established a precise legislative framework pertaining to various sorts of electronic crimes, as well as processes for their investigation and punishment. Any conduct prohibited by the penal laws/laws of the nation is a criminal offence punishable by law. Current criminal tendencies may now be learnt in order to prevent future crimes, making penalties simpler for law enforcement authorities. To reduce horrific crimes in Pakistan, the Serious Crime Act 2007 includes a Serious Crimes Prevention Order. It is a court order designed to protect the public by prohibiting and limiting an individual's involvement in criminal activity. The Artificial Intelligence is now become a very important tool in almost every industry. It is an important part of computer science, and it works as robot works and in a digital environment as information is provided to is. In this age of

technological advancement, the use of technology to resolve our issues of every industry has become sine que none and its use saves the resources and precious times served on doing any task without using modern technology. The rapid changes in the lifestyle of the masses require use of advance gadgets that are compatible with the pace of social, economic and technological changes. Now artificial intelligence is being used in cyber security crimes and hence it opens the door for using serious crimes in the country to nab the criminal's rather white collar criminals. With the advancement of technological fields, the criminals are now become clever and using the technology for committing crimes, therefore burden to track these advanced-technological criminals, is the responsibility of Law enforcement agencies. The artificial intelligence in investigating the sophisticated crimes of cyber security has become inseparable. The law enforcement agencies cannot perform their legal and constitutional duties without any proper legislation on the subject. The legislature is suggested to make law in accordance with law and that one should be compatible with The Serious Crime Act 2007.

7. Conclusions

This study proposes a novel deep learning framework by integrating ResNet-50/34/18 and SRU models for detecting anomaly activities in the UCF-Crime and binary datasets. A few anomalies take place in videos where people cannot be seen, such as car accidents. Besides, other anomaly events happen for a few seconds in a short video (10 s), but most parts of such videos show a normal event. Regardless of stated limitations, the CNN + SRU models outscored other models on the UCF-Crime dataset with 90.20% AUC and 90.19% accuracy (average of three models that are ResNet18 + SRU, ResNet34 + SRU, and ResNet50 + SRU). In addition to the 14 classes of the UCF-Crime dataset, dividing the dataset into two major classes (Ucfcimes and Binary) shows improved results in terms of accuracy and AUC. The focused video frames extracted from the original videos of anomaly events have significantly improved the detection accuracy. Among other CNN models implemented for anomaly events, ResNet-50 combined with SRU provides improved results. With excellent feature extraction, ResNet50 and SRU significantly improved the performance measures. With ResNet50 + SRU, the classifier efficiently detected the anomaly classes for Ucfcimes and Binary datasets. Though the experimental results

demonstrate that CNN + SRU performed better than other related models in terms of accuracy, precision, and AUC, we still look to improve classification of all kinds of anomalies in the UCF-Crime dataset. One of the approaches is to add attention layers to the CNN + SRU as part of the future work. The attention layer can be integrated with CNN and/or SRU. With this approach, the future model can be focused more precisely on the anomaly events in a video. Silent videos can be used to detect anomalies in terms of audio signals since most of the time, only silent video is available. So, if we use audio signal synthesis, we can make video surveillance more effective [53].

Credit author statement

Maryam Qasim: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, **Elena Verdu:** Data curation, Writing – original draft, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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